

News Bias Detection Using Deep Learning Models

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Abstract—Political polarization has amplified the need for automatic tools that showcase media bias and help readers contextualize news coverage. We present a systematic study of various approaches using deep learning for multi-class political bias detection in news articles. Leveraging the QBias dataset of 21,754 U.S. news articles professionally annotated by the news site *AllSides* as *Left*, *Center*, or *Right*, we benchmark three models:

- 1) a lightweight long short-term memory (LSTM) network trained from scratch,
- 2) a term frequency-inverse document frequency logistic-regression model, and
- 3) a fine-tuned BERT model.

Our experiments show that the LSTM and BERT models achieve near-perfect accuracy of 0.9917 and 0.9908, respectively, largely outperforming the TF-IDF’s accuracy of 0.9117. Feature exploration reveals that removing *source* reduces accuracy to 0.47, underscoring the strong correlation between publisher identity and perceived bias. Qualitative analysis of misclassified article show us that politically charged keywords and political figures taken out of context confuse the models and cause incorrect predictions.

Index Terms—LSTM, BERT, TF-IDF, Logistic Regression, News Bias

I. INTRODUCTION

Political bias in news reporting can shape public opinion and exacerbate polarization. Tools that automatically identify bias enable readers to consume diverse viewpoints and improve media literacy. We address this task as a *three-way text classification* problem: given an article, predict whether its perspective leans *Left*, *Center*, or *Right*. Our contributions are three-fold:

- We analyze the QBias dataset, a 21k-article corpus labeled by media-bias experts at *AllSides*.
- We provide strong models across two architectures: recurrent based LSTM and transformer based BERT, along with TF-IDF logistic regression. We also provide an evaluation of models’ predictions.
- We release code and resources that facilitate future research on media-bias detection.

II. PREVIOUS WORK

Previous works have identified that news bias detection is an important task and that deep learning models can identify and classify political bias in news. [1] introduces a dataset of nearly 34,000 news articles labeled by political ideology (left, center, right), and aims to predict article bias without relying on the source. The authors develop two methods using BERT models

to prevent models from learning the media source instead of true bias: Adversarial Media Adaptation and Triplet Loss Pre-training (TLP).

[2] uses LLMs to detect news bias. The research presents a new dataset labeled by GPT-4o for detecting seven types of biases (e.g., political, gender, racial) in news articles. The authors tested five transformer models and found BERT to be the most effective, especially for political and religious bias detection. While class imbalance affected some models’ performance, BERT consistently outperformed others across most categories.

[3] reviews methods for detecting bias in online news using different NLP methods. It examines techniques like feature extraction and classification using models such as BERT, LSTM, and traditional machine learning approaches like logistic regression and Naive Bayes.

III. DATASET

We use the Qbias Dataset from the IR Group, which is a GitHub Repository released in 2023 with pre-collected data from *AllSides*, as well as a scraper for further data collection if necessary. The data was scraped in November 2022. *AllSides* is an American news aggregator that labels articles as left-, right-, or center-leaning based on professional opinions.

AllSides creates its headline roundups by consulting with a multi-partisan expert panel reviews and ratings from thousands of Americans across the political spectrum. Our dataset uses these labels from *AllSides* to classify the political leaning of each news article. As a result, the bias ratings from *AllSides* included in the dataset represent a diverse set of perspectives.

The dataset includes title, tags, headline, source, text, and bias ratings of news articles on *AllSides*. There are 21,754 total samples, of which 10,275 are left-leaning, 7,226 are right-leaning, and 4,253 are center-leaning.

IV. METHODS

A. Pre-processing

We concatenated each article’s *title*, *heading*, *tags*, *source*, and *text* into a single string, and converted all words to lower case. The dataset was stratified into 70% train, 15% validation, and 15% test splits so that the class distribution remained uniform across partitions. For our LSTM, we applied the `bert-base-uncased` WordPiece tokenizer, while the TF-IDF Logistic Regression and fine-tuned BERT models consumed the raw strings.

B. Models

a) *Custom LSTM (Baseline)*: Our lightweight LSTM baseline contains six layers: (i) An input embedding layer, (ii) dropout ($p=0.5$), (iii) a bidirectional LSTM with an L2 kernel regularizer ($\lambda=10^{-3}$), (iv) dropout ($p=0.5$), (v) a dense layer with ReLU activation function with an L2 kernel regularizer ($\lambda=10^{-3}$), (vi) and a 3-way soft-max output. We used the Sparse Categorical Cross-Entropy function with Adam optimizer for twenty epochs on the validation dataset.

b) *TF-IDF Logistic Regression*: To establish a strong TF-IDF model, we built a scikit-learn pipeline. The vectorizer extracts uni- and bi-gram features, caps the vocabulary at 5,000, and removes standard English stop-words. The classifier is trained until convergence ($\text{max_iter} = 1000$) with the L2 regularized LBFGS optimizer. Class probabilities are obtained directly from the softmax output layer of the logistic model.

c) *BERT Fine-Tuning*: For our transformer approach, we fine-tuned the `bert_medium_en_uncased` preset provided by Keras. We first attach a randomly initialized classification head (`num_classes=3`) and perform a quick full-model warm-up. Next, we re-compile with an explicit `SparseCategoricalCrossentropy` loss, the Adam optimizer, and `jit_compile=True`. To stabilize training, we freeze the backbone and fine-tune only the classifier for ten additional epochs using a batch size of 32.

V. EVALUATION

After training our three models, we produced classification reports for each model. We analyze metrics such as accuracy, precision, recall, and F-1 Scores. For each of our models, we created a confusion matrix with a heatmap to visualize the correct and incorrect classifications. We then analyzed the softmax probability distributions to understand the confidence of our models. We also examined fifteen incorrect predictions text samples to understand why some of them were being classified incorrectly with relatively high confidence scores.

VI. RESULTS

For model input and output, we mapped the labels "left", "center", and "right" as 0, 1, and 2 respectively. Thus, our tables and figures below will reflect this labeling.

Table I shows the classification report for our custom LSTM. The model achieved a 99.17% accuracy. We note that our custom LSTM model performs extraordinarily well for all classes, proving to be a strong baseline model. Both precision and recall are close to 1.0 across all classes, indicating that the model rarely confuses one class for another. Fig. 1 shows the confusion matrix for the LSTM. The figure shows that the model correctly classified nearly all instances of our test dataset, indicating that our model generalizes very well.

TABLE I
LSTM CLASSIFICATION REPORT

	Precision	Recall	f1-score
0	1.00	0.99	0.99
1	0.98	1.00	0.99
2	0.99	0.99	0.99

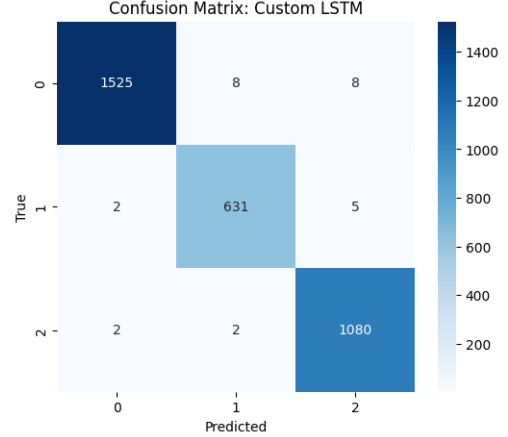


Fig. 1. LSTM confusion matrix.

TABLE II
TF-IDF LOGISTIC REGRESSION CLASSIFICATION REPORT

	Precision	Recall	f1-score
0	0.89	0.96	0.93
1	0.92	0.80	0.85
2	0.93	0.91	0.92

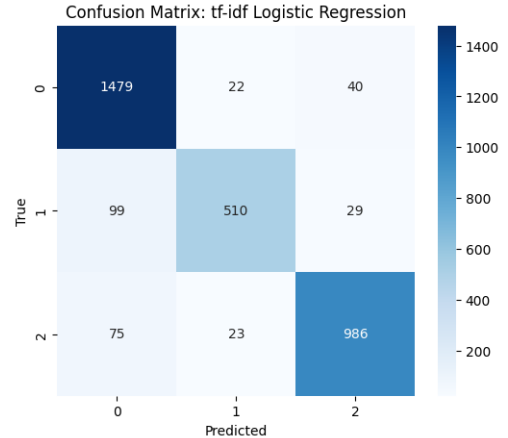


Fig. 2. TF-IDF Logistic Regression confusion matrix.

Table II shows the classification report for our Logistic Regression model trained on TF-IDF vectorized features. The model achieved a 91.17% accuracy. In the confusion matrix in Fig. 2, we more clearly visualize that the model struggles to properly classify center-leaning articles. Unlike an LSTM

which uses sequential data to learn patterns in word order, TF-IDF treats words as independent and does not capture word order. This lack of context might be the cause of the poorer performance of the TF-IDF logistic regression model.

TABLE III
FINE-TUNED BERT CLASSIFICATION REPORT

	Precision	Recall	f1-score
0	1.00	0.99	0.99
1	0.98	0.99	0.99
2	0.99	0.99	0.99

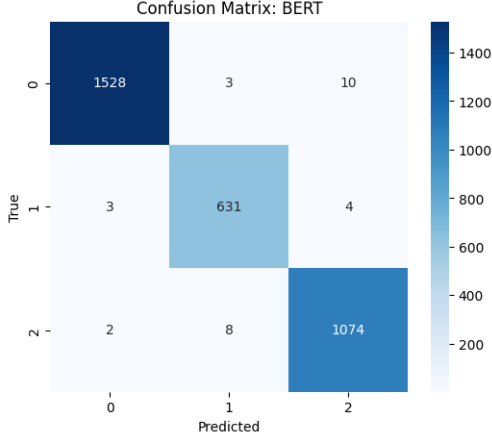


Fig. 3. Fine-Tuned BERT confusion matrix.

Table III shows our fine-tuned BERT classification report. The fine-tuned BERT model performs well for all classes, similar to the custom LSTM baseline model. It achieved an accuracy of 99.08%. Fig. 3, the confusion matrix for BERT, shows that the model has very few incorrect predictions on the test set. Our fine-tuning process yielded a high-performing model that generalized well.

Furthermore, when we tested our fine-tuned BERT model on current news (outside of the dataset), the model performed perfectly, correctly classifying left, center, and right leaning articles. The news articles we tested on were all related to the same topic with the same keywords, but had a different source and main text. Despite being fairly similar, the model was still able to correctly classify the political bias of the articles, emphasizing the ability of the model to generalize well.

VII. DISCUSSION

a) Source identity dominates performance: When training our model, we found that including the `source` field dramatically improved the performance of all three of our models. For example, when we exclude it from training, our F-1 drops from 0.99 to 0.47 for the fine-tuned BERT model. This confirms that U.S. news outlets possess highly distinctive stylistic and topical footprints that reveal their political biases. Although useful for maximizing accuracy, using the `source` field in training does potentially risk reinforcing preexisting

assumptions about a publisher’s slant and may not transfer to unseen outlets or international contexts, as [1] suggested.

b) TF-IDF Logistic Regression struggles with cross-class keywords: The TF-IDF Logistic Regression model performs competitively overall (accuracy = 0.91) but frequently confuses articles that mention partisan entities. For instance, the mere appearance of “*Fox News*” in a piece led to a highly confident “Right-leaning” prediction. Because TF-IDF treats n -grams in isolation, it cannot exploit broader discourse cues that might signal disagreement or irony.

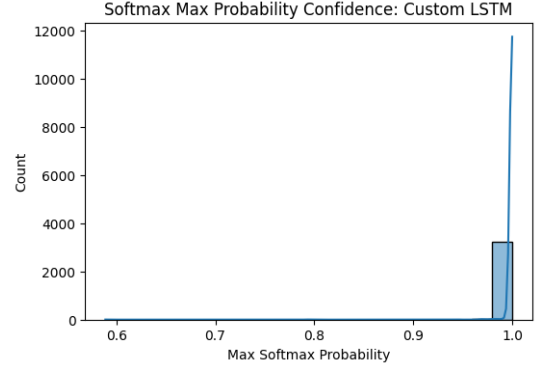


Fig. 4. LSTM Confidence Scores

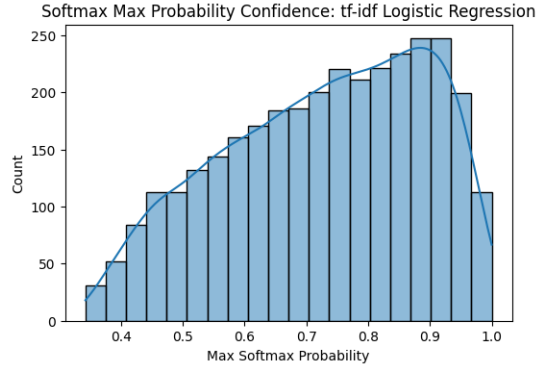


Fig. 5. TF-IDF Logistic Regression Confidence Scores.

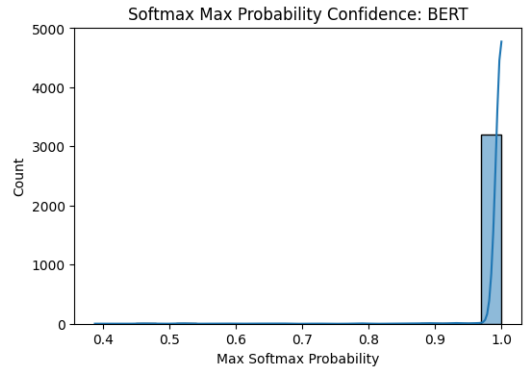


Fig. 6. Fine-Tuned BERT Confidence Scores.

c) *Neural network models are accurate yet over-confident*: Fig. 4 and Fig. 6 show that both the LSTM baseline and fine-tuned BERT assign probabilities close to 1.0 for the vast majority of test instances, whereas TF-IDF (Fig. 5) exhibits a much more spread out distribution. Although producing few incorrect classifications, the LSTM model is highly confident even for most misclassified predictions. Our fine-tuned BERT model, on the other hand, has far lower confidence for its incorrect predictions. While high confidence is reassuring when the prediction is correct, out-of-domain articles could trigger confident but incorrect classifications, possibly undermining user trust. Temperature scaling or mixup training are possible remedies we leave for future work.

d) *Limitations and next steps*: First, QBias covers only U.S. outlets; generalizing beyond this small portion of global news remains an open task. Second, the current data includes both factual reporting and opinion pieces. Future work could isolate one or the other. Extending the dataset temporally, across languages, and with finer-grained bias dimensions (e.g. economic vs. social issues) such as in [2] would potentially produce a more nuanced model. Our final model could be deployed as a browser extension that automatically classifies political bias of a given news article.

VIII. SUMMARY

- **Source matters most**: the `source` field alone drives significant predictive power. Our models appear to primarily learn from the source name instead of the article text itself.
- **All models work well, but...** neural network models (LSTM and BERT) achieve F-1 of around 0.99, vastly outperforming a strong TF-IDF model (0.91). However, these models might be over-confident.
- **Lightweight wins on efficiency**: our custom LSTM rivals BERT while requiring far less compute, making it practical for real-time or on-device deployment.
- **User impact**: integrating such models as a browser extension can nudge readers toward a more diverse set of viewpoints and raise awareness of hidden partisan cues.

Future work should explore debiasing strategies that mask source identifiers, generalize to unseen sources, and richer media-bias taxonomies that go beyond the Left-Center-Right triad.

IX. REFERENCES

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